Practice workbook · Scale a rate per 100 000 people to a whole population

Disease cases or deaths are given, often read from a graph, per 100 000 people; the number in a stated population is required.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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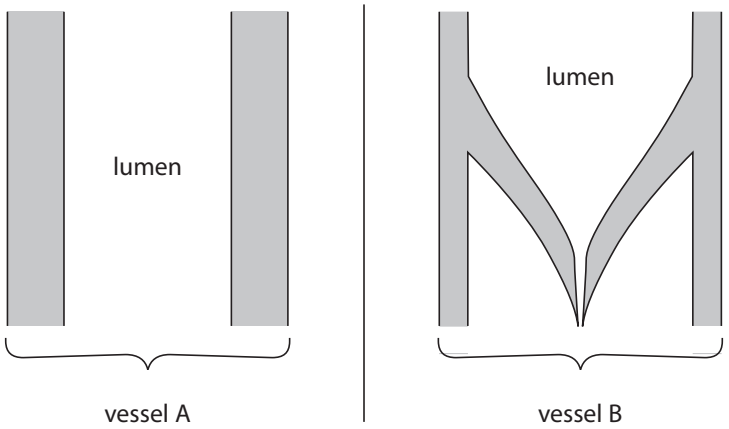
Type ID: HBI-MATH-RATE-PER-100000-SCALING

1. Question 3 · part b.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part b.ii only.

3 (a) The diagram shows two types of blood vessel, A and B.



(i) Draw an arrow on the diagram to show the direction of blood flow in vessel B.

(1)

(ii) Identify each type of blood vessel, giving explanations for why there are differences in their structures.

(4)

Vessel A

Vessel B

(iii) State the name of one other type of blood vessel.

(1)

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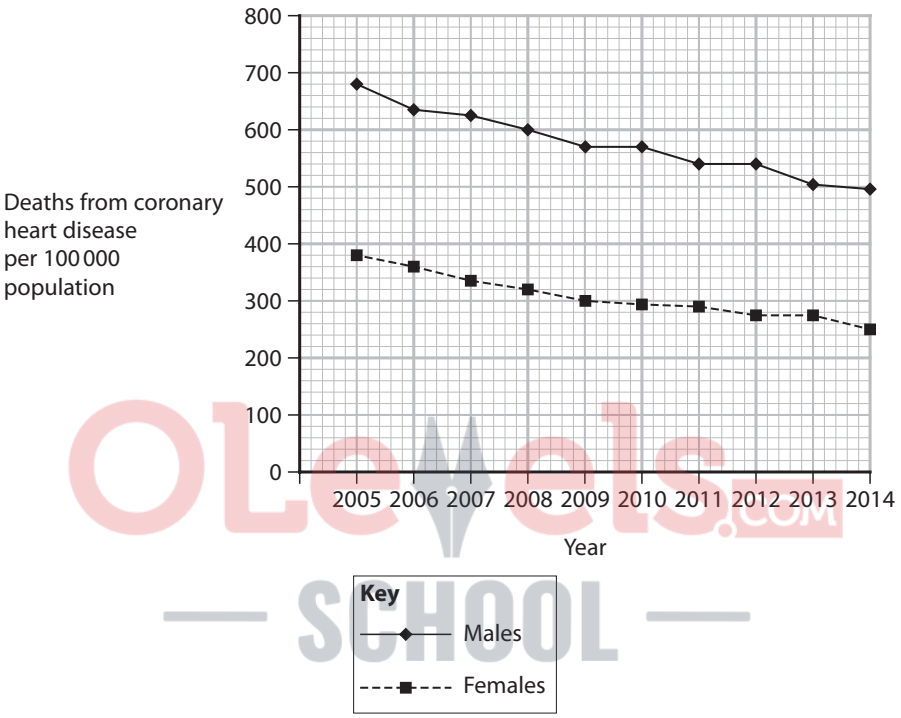
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1. Question 3 · part b.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part b.ii only.

(b) Coronary heart disease kills many people each year. The graph shows the number of deaths per 100 000 of the whole population in a part of the world.



(i) Describe the trends shown by the graph.

(3)

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1. Question 3 · part b.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part b.ii only.

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(ii) The part of the world for which the data was collected had a population of 60 million people in 2006.

Calculate the number of men who died from coronary heart disease in 2006.

(3)

number of men who died = ...

(Total for Question 3 = 12 marks)

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Working and answer

2. Question 6 · part a.iii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iii only.

6 *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

Age group in years	Number of deaths in one year per million people			
	Males		Females	
	S. aureus	**MRSA**	***S. aureus***	**MRSA**
under 45	1.8	0.5	1.4	0.4
45–54	6.0	2.5	4.3	2.4
55–64	15.7	8.2	8.1	4.2
65–74	41.2	26.1	21.9	12.5
75–84	128.6	86.8	81.1	51.6
85 and over	416.6	299.9	200.3	146.4

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

(2)

1 .....

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2 .....

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2. Question 6 · part a.iii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iii only.

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(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

(2)

ratio = ...

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(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

(2)

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number = ...

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(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

(2)

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2. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

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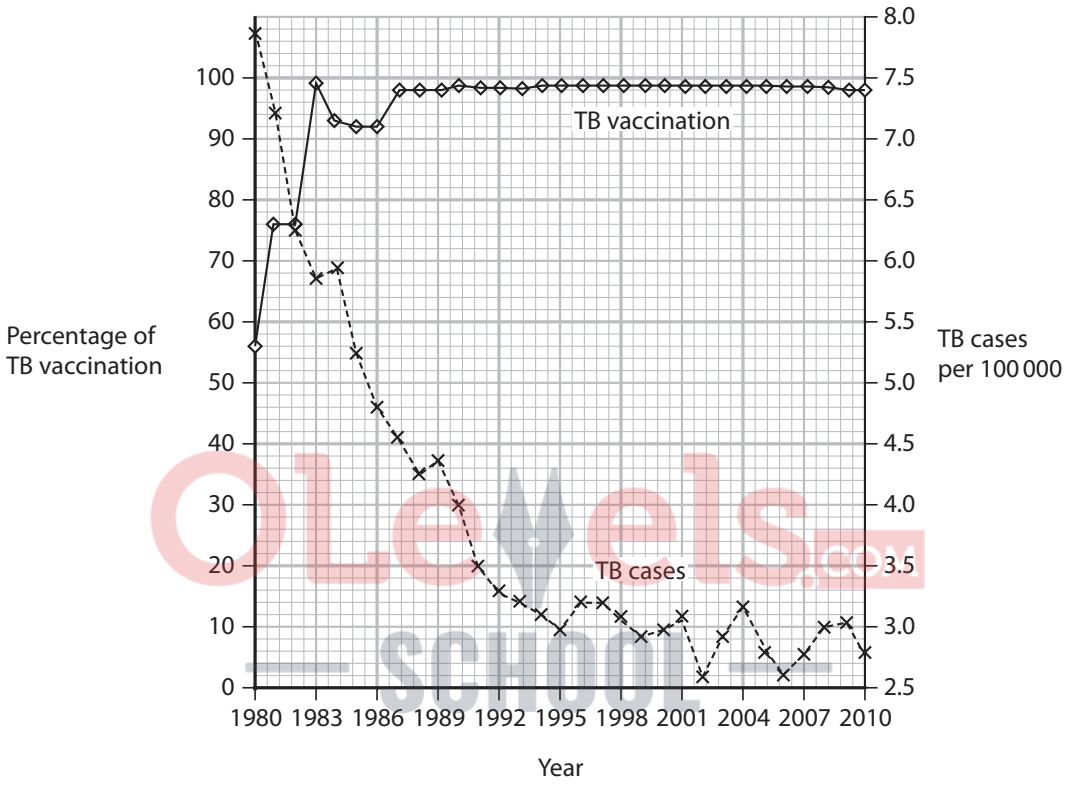
3. Question 4 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

4 The graph shows the percentage of people vaccinated against tuberculosis (TB) in a country between 1980 and 2010.

It also shows the number of cases of tuberculosis in that country.



(a) (i) Tuberculosis is caused by the same type of organism as typhoid.

What is the type of organism?

(1)

- ☒ **A** bacterium
- ☒ **B** fungus
- ☒ **C** protozoan
- ☒ **D** virus

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3. Question 4 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

(ii) Describe the relationship between the percentage vaccination and the number of cases of TB.

(4)

(b) The population of this country in 2010 was 3.45 million.

Calculate the number of cases of tuberculosis in this country in 2010.

(3)

number of cases = ...

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3. Question 4 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

(c) Explain how vaccination can give a person immunity to a disease.

(5)

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(Total for Question 4 = 13 marks)

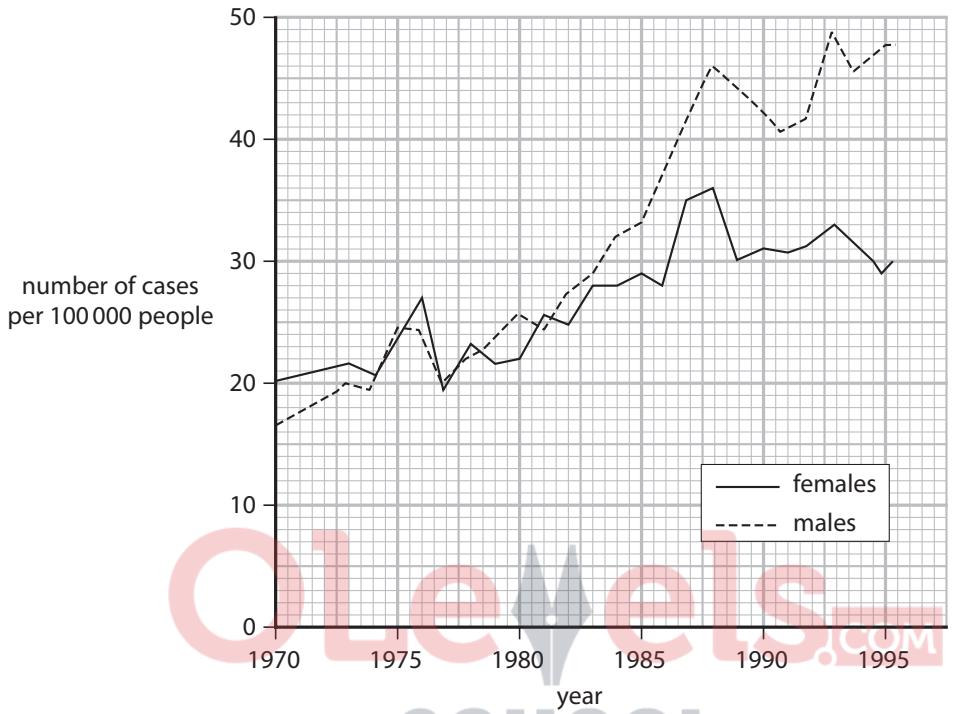
Working and answer

4. Question 6 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

6 A form of skin cancer called melanoma has been increasing in both males and females.



(a) (i) Describe the trends shown by the graph.

(3)

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4. Question 6 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Suggest why cancer can occur in the skin.

(2)

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(b) (i) Calculate the mean increase in the number of melanoma cases per 100 000 males per year between 1970 and 1995.

(3)

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mean increase per 100 000 males per year = ...

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(ii) In 1970, the population of this area of the world was 7 million.

Calculate the number of females who had melanoma.

(3)

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number of females with melanoma = ...

(Total for Question 6 = 11 marks)

4. Question 6 · part b.ii · 3 marks · Working and answer

Working and answer

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5. Question 4 · part c.ii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.ii only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

(2)

diarrhoea

malaria

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5. Question 4 · part c.ii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.ii only.

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(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

(1)

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

(2)

elderly people

indoor spaces

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(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

(2)

1 .....

2 .....

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

(3)

number of people = ...

5. Question 4 · part c.ii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.ii only.

(iii) Calculate the ratio of the number of people with non-infectious diseases in a higher income country to the number of people with non-infectious diseases in a lower income country (lines 20 and 21).

Give your answer in the form n : 1

(3)

ratio = ... : .....

(iv) Suggest why there are more cases of dementia and cancers in higher income countries (lines 21 and 22).

(2)

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(Total for Question 4 = 15 marks)

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5. Question 4 · part c.ii · 3 marks · Working and answer

Working and answer

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